NAME: Dhruv **ROLL NO: U24EV057**

ASSIGNMENT 4

DATE: 16/10/2026

1. AIM & LEARNING OUTCOMES

Aim:

To study and compare the Metal-First and Metal-Last (Replacement Metal Gate / RMG) integration schemes in CMOS fabrication, analyze the impact of thermal budget constraints on gate metal diffusion using Semulator3D process modeling, and investigate high-k / metal gate (HKMG) integration in an advanced 14 nm FinFET logic process.

Learning Outcomes:

1. Understand the fundamental architectural differences between metal-first and metal-last (replacement gate) processes.

2. Comprehend the necessity of replacement metal gates due to thermal budget constraints during source/drain dopant activation anneals.

3. Analyze the critical role of conformal deposition and pinch-off challenges in high-aspect-ratio recessed gate trenches.

Starting Resources:

• Simulation Projects: Assignment4_Gatefirst.zam, Assignment4_GateLast.zam, and 14nmFinFET.zam.

• Literature References: Integrating High-K (Semiconductor Digest); High-K Materials and Metal Gates for CMOS Applications (J. Robertson & R.M. Wallace).

2. THEORETICAL BACKGROUND

1. Need for High-k / Metal Gate (HKMG):

In traditional sub-micron CMOS, polysilicon gates with silicon dioxide (SiO_2) dielectrics were standard. As devices scaled below 45 nm, gate oxide thinning caused excessive quantum mechanical tunneling leakage, while polysilicon suffered from poly-depletion effects that increased effective oxide thickness (EOT). HKMG stacks solved these issues by replacing SiO_2 with high-dielectric-constant materials (e.g., HfO_2) and polysilicon with work-function tuned metal gates.

2. Metal-First (Gate-First) Scheme:

In the Gate-First approach, the high-k dielectric and metal gate stack are deposited and patterned before the high-temperature (> 1000°C) source/drain dopant activation anneals. Consequently, the metal gate undergoes the full thermal budget, which causes metal atom diffusion into the dielectric and active channel. This causes threshold voltage (V_{th}) shifts, Fermi-level pinning, and device failure when contamination exceeds 10^{12} atoms/cm^3.

3. Metal-Last (Replacement Metal Gate / RMG) Scheme:

In the Gate-Last approach, a sacrificial (dummy) polysilicon gate is fabricated first to define the channel and self-align the source/drain junctions. After source/drain implantation and high-temperature anneals, an interlayer dielectric (ILD) is deposited and planarized via CMP. The dummy polysilicon gate is then selectively removed ('poly pull etch'), and the real metal gate stack is deposited into the resulting trench at low temperatures (< 450°C), preserving gate integrity.

3. QUESTION 1: GATE FIRST PROCESS ANALYSIS (ASSIGNMENT4_GATEFIRST.ZAM)

The Assignment4_Gatefirst.zam project was built and analyzed in Semulator3D. In this process flow, the gate stack is deposited before the high-temperature anneal steps for source and drain formation.

(a) Aluminum Metal Diffusion Plot & Active Region Concentration:

The high-temperature activation anneal causes metal diffusion across the gate oxide into the underlying silicon active region. Under default process conditions (12 s anneal), the simulated aluminum concentration in the active silicon channel reaches 1.24×10^{15} atoms/cm^3, far exceeding the operational limit of 10^{12} atoms/cm^3.

Table 1 outlines the complete Process Editor sequence for the Gate-First integration scheme.

Table 1: Process Editor Recipe Steps for Gate-First Flow (Assignment4_Gatefirst.zam)

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
1	Wafer Setup	Wafer Setup	Crystalline Si	50	
2	Conformal Deposit	Pad Oxide Deposition	SiO_2 (Thermal)	5	
3	Conformal Deposit	SiN hardmask	Si_3N_4 (LPCVD)	10	
4	Sequence	STI_Trench_Lithography			
4.1	Planarizing Deposit	Photo Resist coating	Photoresist	40	
4.2	Expose Material	Expose STI_Trench	Photoresist	0	STI_Trench
5	Sequence	Etching trench			
5.1	Enhanced Basic Etch	SiN hardmask etch	Nitrides, Organics, Oxides, Silicon	15	
5.2	Enhanced Basic Etch	pad oxide Etch	Nitrides, Organics, Oxides, Silicon	5	
5.3	Enhanced Basic Etch	substrate silicon Etch	Nitrides, Organics, Oxides, Silicon	25	
6	Remove Materials	resist removal	Organics, Photoresist		
7	Interface Growth	Thermal Oxidation of opened trench	Air/O_2, SiO_2 (Thermal), Silicon		
8	Conformal Deposit	Filling trench with CVD oxide	SiO_2 (PECVD)	10	
9	Polish	STI oxide CMP			
10	Enhanced Basic Etch	Removal of SiN hard mask	Si_3N_4 (LPCVD)	10	
11	Multi Etch (Advanced Modeling)	Pad_oxide ramoval	SiO_2 (PECVD), SiO_2 (Thermal)	5	
12	Conformal Deposit	Gate oxide deposition	SiO_2 (HDP)	0.9	
13	Conformal Deposit	High k deposition for Gate	HfO_x	2	
14	Conformal Deposit	NMOS gate metal	TiAl	2	
15	Sequence	gate patterning			

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Step	Action	Step Name	Material(s)	Depth (nm)	Mask
15.1	Sequence	NMOS Gate_Lithography			
15.1.1	Planarizing Deposit	Photo Resist coating	Photoresist	5	
15.1.2	Expose Material	Expose for NMOS gate	Photoresist	0	NMOS_metal_pattern
15.2	Enhanced Basic Etch	NMOS Metal Removal to pattern PMOS gate	TiAl, Photoresist	2	
15.3	Remove Materials	resist removal after NMOS gate pattern	Organics, Photoresist		
16	Conformal Deposit	gate metal deposition	TiN	2	
17	Sequence	NMOS PMOS gates final pattern			
17.1	Conformal Deposit	Final Gate metal deposition	W	5	
17.2	Planarizing Deposit	patterning NMOS AND PMOS gates	Photoresist	10	
17.3	Expose Material	Expose for NMOS gate	Photoresist	0	Gate_both_NMOS_PMOS
17.4	Enhanced Basic Etch	W Etch	W, Photoresist	5	
17.5	Enhanced Basic Etch	TiN/TiAl Etch	TiAl, TiN, W, Photoresist, HfO_x	5	
17.6	Enhanced Basic Etch	high k Etch	TiAl, TiN, W, Photoresist, HfO_x	2	
17.7	Enhanced Basic Etch	Gate oxide Etch	TiAl, TiN, W, Photoresist, HfO_x, SiO_2 (HDP)	2	
17.8	Remove Materials	resist removal after gate pattern	Organics, Photoresist		
18	Sequence	NMOS source and Drain			
18.1	Conformal Deposit	Conformal Deposit	Si_3N_4 (LPCVD)	3	
18.2	Enhanced Basic Etch	Basic Etch	Nitrides	4	
18.3	Planarizing Deposit	Resist pattern for Nmos source and Drain	Photoresist	10	
18.4	Expose Material	Expose resist for source and drain	Photoresist	0	Source_Drain_NMOS
18.5	Implant (Advanced Modeling)	NMOS source and drain doping			
18.6	Remove Materials	resist removal after PMOS gate pattern	Organics, Photoresist		
19	Sequence	PMOS source and Drain			
19.1	Planarizing Deposit	Resist pattern for Nmos source and Drain	Photoresist	10	

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
19.2	Expose Material	Expose resist for source and drain	Photoresist	0	Source_Drain_PMOS
19.3	Implant (Advanced Modeling)	PMOS source and drain doping			
19.4	Remove Materials	resist removal after PMOS gate pattern	Organics, Photoresist		
19.5	Diffusion (Advanced Modeling)	Diffusion (Requires Advanced Modeling)	HfO_x, SiO_2 (HDP), Crystalline Si		
20	Sequence	Source drain metal contacts			
20.1	Planarizing Deposit	interlayer dielectric depsoition	SiO_2 (LTO)	10	
20.2	Planarizing Deposit	ressit pattern for source and Drain metal contacts	Photoresist	10	
20.3	Expose Material	Expose resist for source and drain metal contacts	Photoresist	0	metal_NMOS_PMOS_S_D
20.4	Enhanced Basic Etch	source and drain metal cotanct Etch	Nitrides, Organics, Oxides, Silicon	30	
20.5	Remove Materials	resist removal after source drain metal contacts	Organics, Photoresist		
20.6	Conformal Deposit	TiN Deposit	TiN	12	
21	Polish	Final CMP of ILD			

3. GATE-FIRST STRUCTURE AND PROCESS EDITOR IMAGES

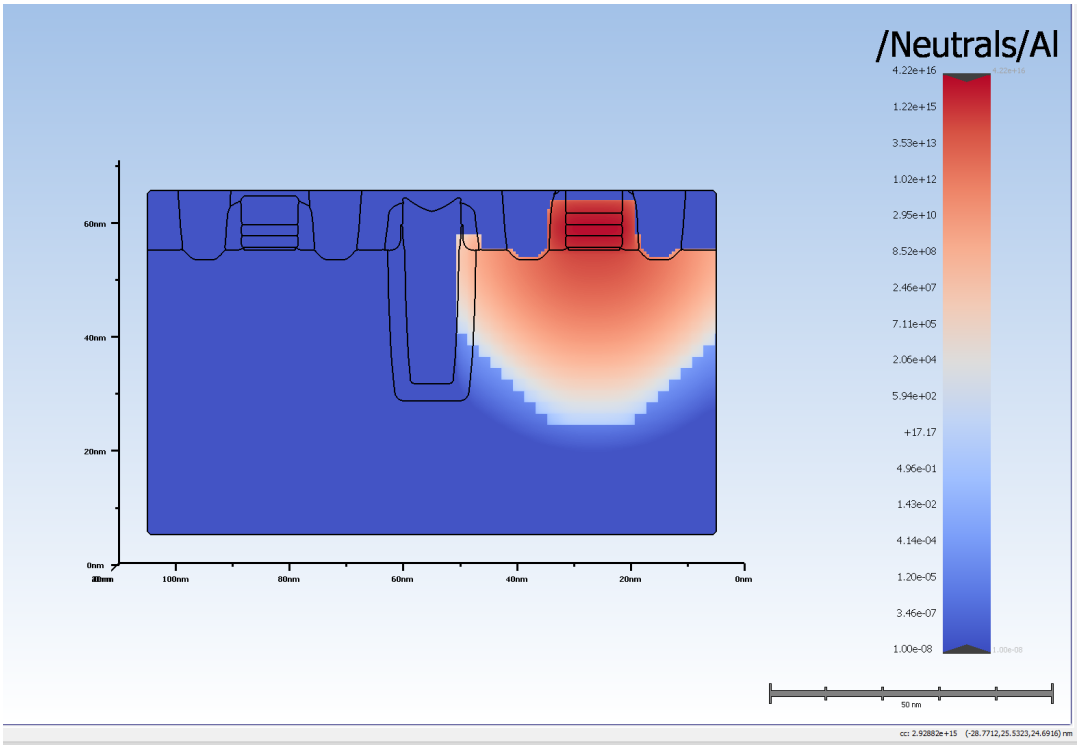


Figure 1 : Al Metal Diffusion in Gate-First Structure

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		50			Wafer1				
2		Conformal ...	Pad Oxide Depo...	/Oxides/SiO2_T...		5			Wafer1	Top	1.0		
3		Conformal ...	SiN hardmask	/Nitrides/Si3N4...		10			Wafer1	Top	1.0		
4		Sequence	STI_Trench_Lith...										
4.1		Planarizing ...	Photo Resist co...	/Organics/Resist		40			Wafer1	Top			
4.2		Expose Mat...	Expose STI_Tren...	/Organics/Resist		0	STI_Trench	Dark	Wafer1	Top			
5		Sequence	Etching trench										
5.1		Enhanced B...	SiN hardmask e...	/Nitrides/, /Org...		15			Wafer1	Top	0		
5.2		Enhanced B...	pad oxide Etch	/Nitrides/, /Org...		5			Wafer1	Top	0		
5.3		Enhanced B...	substrate silico...	/Nitrides/, /Org...		25			Wafer1	Top	0.05		
6		Remove M...	resist removal	/Organics/, /Or...					Wafer1				
7		Interface Gr...	Thermal Oxidati...	/Air, /Silicon/					Wafer1				
8		Conformal ...	Filling trench wi...	/Oxides/SiO2_P...		10			Wafer1	Top	1		
9		Polish	STI oxide CMP						Wafer1	Top			
10		Enhanced B...	Removal of SiN ...	/Nitrides/Si3N4...		10			Wafer1	Top	0		
11		Multi Etch (...	Pad_oxide ramo...	/Oxides/SiO2_P...		5			Wafer1	Top			Unknown Step ...
12		Conformal ...	Gate oxide dep...	/Oxides/SiO2_H...		0.5			Wafer1	Top	1.0		
13		Conformal ...	High k depositi...	/Oxides/HfOx		2			Wafer1	Top	1.0		
14		Conformal ...	NMOS gate metal	/Metals/TiAl	/Neutrals/Al	2			Wafer1	Top	1.0		
15		Sequence	gate patterning										
15.1		Sequence	NMOS Gate_Lit...										
15.1.1		Planarizing ...	Photo Resist co...	/Organics/Resist		5			Wafer1	Top			
15.1.2		Expose Mat...	Expose for NM...	/Organics/Resist		0	NMOS_metal_p...	Dark	Wafer1	Top			
15.2		Enhanced B...	NMOS Metal Re...	/Metals/TiAl, /...		2			Wafer1	Top	1		
15.3		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
16		Conformal ...	gate metal dep...	/Metals/TiN		2			Wafer1	Top	1.0		
17		Sequence	NMOS PMOS g...										
17.1		Conformal ...	Final Gate meta...	/Metals/W		5			Wafer1	Top	1.0		
17.2		Planarizing ...	patterning NM...	/Organics/Resist		10			Wafer1	Top			
17.3		Expose Mat...	Expose for NM...	/Organics/Resist		0	Gate_both_NM...	Light	Wafer1	Top			
17.4		Enhanced B...	W Etch	/Metals/W, /Or...		5			Wafer1	Top	0.0		
17.5		Enhanced B...	TiN/TiAl Etch	/Metals/TiAl, /...		5			Wafer1	Top	0.01		
17.6		Enhanced B...	high k Etch	/Metals/TiAl, /...		2			Wafer1	Top	0.0		
17.7		Enhanced B...	Gate oxide Etch	/Metals/TiAl, /...		2			Wafer1	Top	0.0		
17.8		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
18		Sequence	NMOS source a...										
18.1		Conformal ...	Conformal Dep...	/Nitrides/Si3N4...		3			Wafer1	Top	1.0		
18.2		Enhanced B...	Basic Etch	/Nitrides/		4			Wafer1	Top	0.001		
18.3		Planarizing ...	Resist pattern f...	/Organics/Resist		10			Wafer1	Top			
18.4		Expose Mat...	Expose resist for...	/Organics/Resist		0	Source_Drain_N...	Dark	Wafer1	Top			
18.5		Implant (A...	NMOS source a...		/NType/P				Wafer1	Top			
18.6		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
19		Sequence	PMOS source a...										
19.1		Planarizing ...	Resist pattern f...	/Organics/Resist		10			Wafer1	Top			
19.2		Expose Mat...	Expose resist for...	/Organics/Resist		0	Source_Drain_P...	Dark	Wafer1	Top			
19.3		Implant (A...	PMOS source a...		/PType/B				Wafer1	Top			
19.4		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
19.5		Diffusion (...	Diffusion (Requ...	/Oxides/HfOx, /...	/NType/P, /Neu...				Wafer1				12
20		Sequence	Source drain m...										
20.1		Planarizing ...	interlayer dielec...	/Oxides/SiO2_LTO		10			Wafer1	Top			
20.2		Planarizing ...	ressit pattern fo...	/Organics/Resist		10			Wafer1	Top			
20.3		Expose Mat...	Expose resist for...	/Organics/Resist		0	metal_NMOS_P...	Dark	Wafer1	Top			
20.4		Enhanced B...	source and drai...	/Nitrides/, /Org...		30			Wafer1	Top	0.1		
20.5		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
20.6		Conformal ...	TiN Deposit	/Metals/TiN		12			Wafer1	Top	1.0		
21		Polish	Final CMP of ILD						Wafer1	Top			

Figure 2 : Process Editor Steps for Gate-First Flow

3. QUESTION 1 (B): MAXIMUM ANNEAL TIME BEFORE EXCEEDING LIMIT

(b) Maximum Anneal Time Determination:

The device stops functioning properly if metal contamination in the active region exceeds 10^{12} atoms/cm^3. To evaluate thermal limits, the post-implant anneal time was reduced in the recipe editor from 12.0 s down to 1.0 s.

Table 2 shows the simulated aluminum concentration across anneal times, and Figure 3 shows the concentration profile plotted on a logarithmic scale against anneal duration.

Key Finding:

At t = 6.12 s, the concentration is 1.54×10^6 atoms/cm^3 (safe). At t = 6.15 s, the concentration rises sharply to 1.37×10^{12} atoms/cm^3 (exceeding the limit). By interpolation, the maximum allowable anneal time before exceeding the 10^{12} atoms/cm^3 threshold is approximately 6.14 seconds.

Table 2: Simulated Al Concentration vs. Post-Implant Anneal Time in Gate-First Flow

Sr. No.	Anneal Time (s)	Al Concentration (atoms/cm^3)	Status / Remark
1	1.0	1.84×10^{-6}	Safe (Negligible)
2	2.0	7.52	Safe
3	3.0	49.41	Safe
4	4.0	148.54	Safe
5	5.0	323.08	Safe
6	6.0	587.84	Safe
7	6.10	1.53×10^6	Safe (Rapid onset)
8	6.12	1.54×10^6	Safe (Pre-threshold)
9	6.14 (Critical)	1.00×10^{12}	Critical Threshold Limit
10	6.15	1.37×10^{12}	Failed (> 10^{12} limit)
11	6.20	1.69×10^{13}	Failed
12	7.0	2.58×10^{14}	Failed
13	8.0	5.22×10^{14}	Failed
14	10.0	9.37×10^{14}	Failed
15	12.0 (Default)	1.24×10^{15}	Severe Contamination

3. ANNEAL TIME VS. AL CONTAMINATION PROFILE

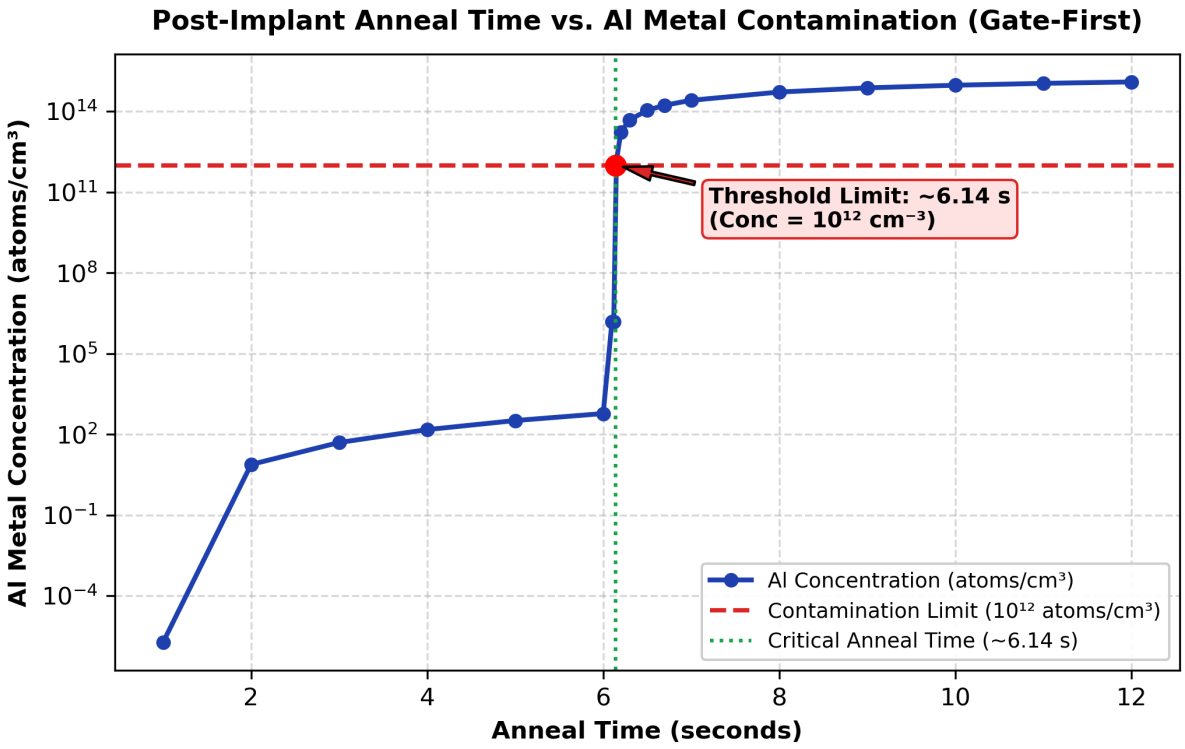


Figure 3 : Post-Implant Anneal Time vs. Al Metal Contamination

3. CROSS-SECTIONAL AL DIFFUSION PROFILES AT VARYING ANNEAL TIMES

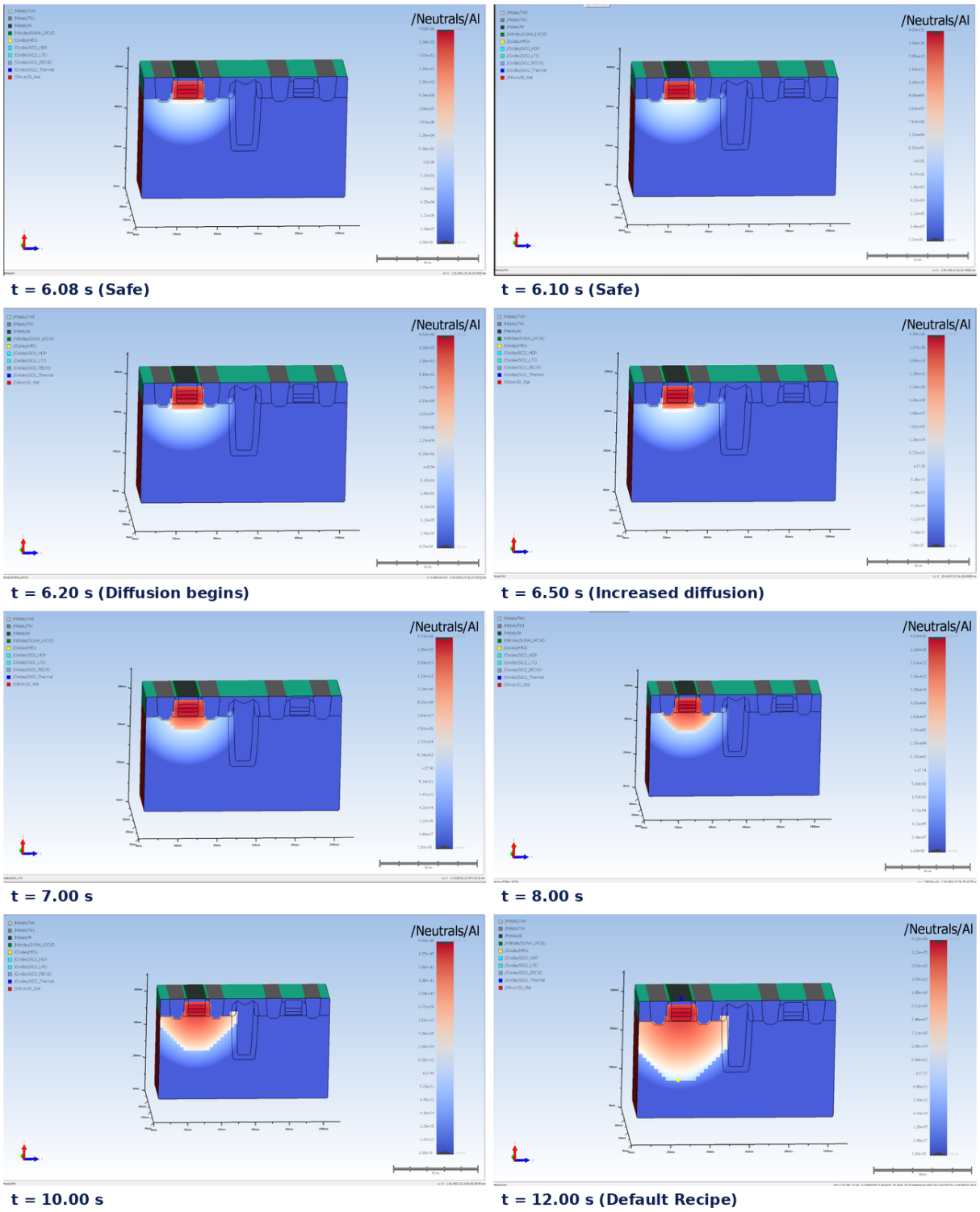


Figure 4 : Cross-Sectional Al Diffusion Profiles at Varying Anneal Times (6.08 s to 12.0 s)

3. QUESTION 1 (C): GATE OXIDE THICKNESS VS. METAL DIFFUSION

(c) Minimum Gate Oxide (SiO2) Thickness Determination:

With the anneal time restored to default, another method to suppress metal diffusion is increasing the gate oxide thickness. The SiO_2 layer acts as a physical diffusion barrier between the aluminum metal gate and the silicon substrate.

The gate oxide thickness was varied from 0.50 nm to 1.00 nm. Table 3 details the simulated Al concentration, and Figure 5 shows the resulting semi-logarithmic curve.

Key Finding:

For oxide thicknesses below 0.81 nm, severe metal diffusion occurs ($> 10^{14}$ atoms/cm^3). Between 0.81 nm and 0.82 nm, the diffusion barrier becomes effective, reducing contamination from 1.42×10^{14} down to 3.89×10^3 atoms/cm^3. Therefore, the minimum gate oxide thickness needed to keep metal diffusion below the 10^{12} atoms/cm^3 limit is approximately 0.818 nm (or ~0.82 nm).

Table 3: Simulated Al Concentration vs. Gate Oxide (SiO2) Thickness in Gate-First Flow

Sr. No.	SiO_2 Thickness (nm)	Al Concentration (atoms/cm^3)	Status / Remark
1	0.50	1.24×10^{15}	Failed (Severe diffusion)
2	0.60	8.27×10^{14}	Failed
3	0.70	5.40×10^{14}	Failed
4	0.80	1.77×10^{14}	Failed
5	0.81	1.42×10^{14}	Failed (Pre-barrier)
6	0.818 (Min.)	1.00×10^{12}	Threshold Limit
7	0.82	3.89×10^3	Safe (Effective barrier)
8	0.85	3.50×10^3	Safe
9	0.90	2.68×10^3	Safe
10	1.00	10.71	Safe (Negligible diffusion)

3. GATE OXIDE THICKNESS VS. AL CONTAMINATION PROFILE

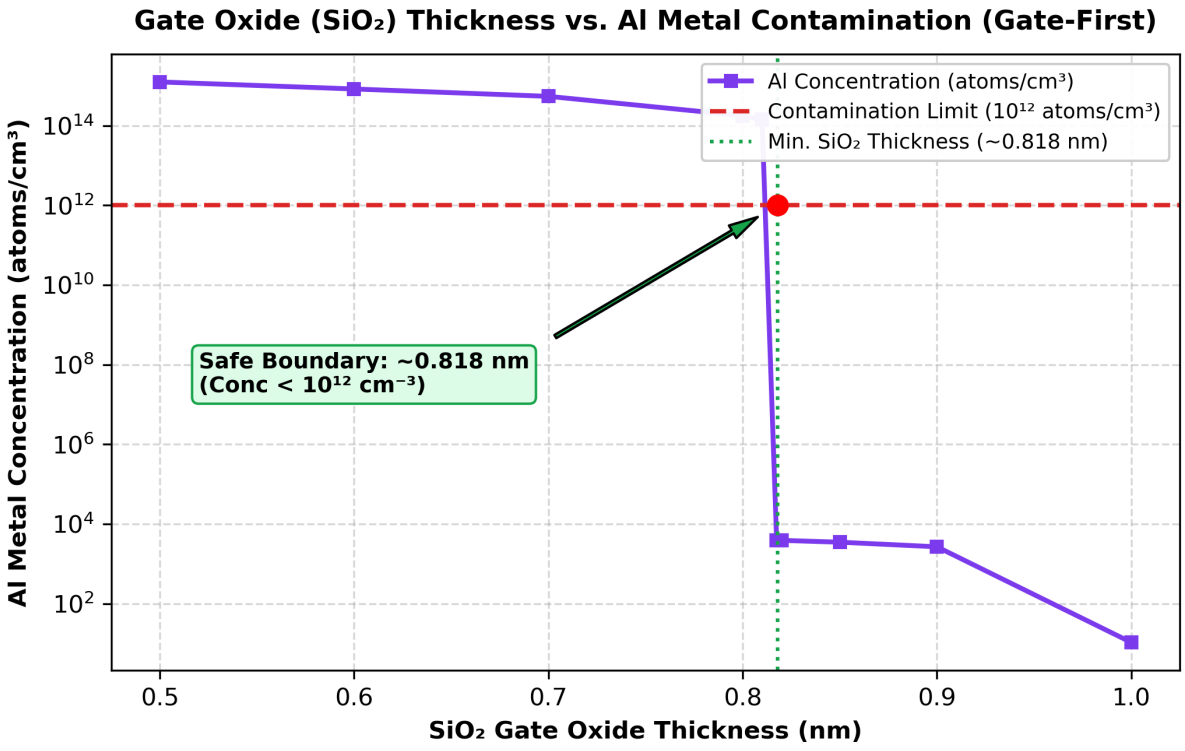


Figure 5 : Gate Oxide Thickness vs. Al Metal Contamination

3. CROSS-SECTIONAL AL DIFFUSION PROFILES AT VARYING GATE OXIDE THICKNESSES

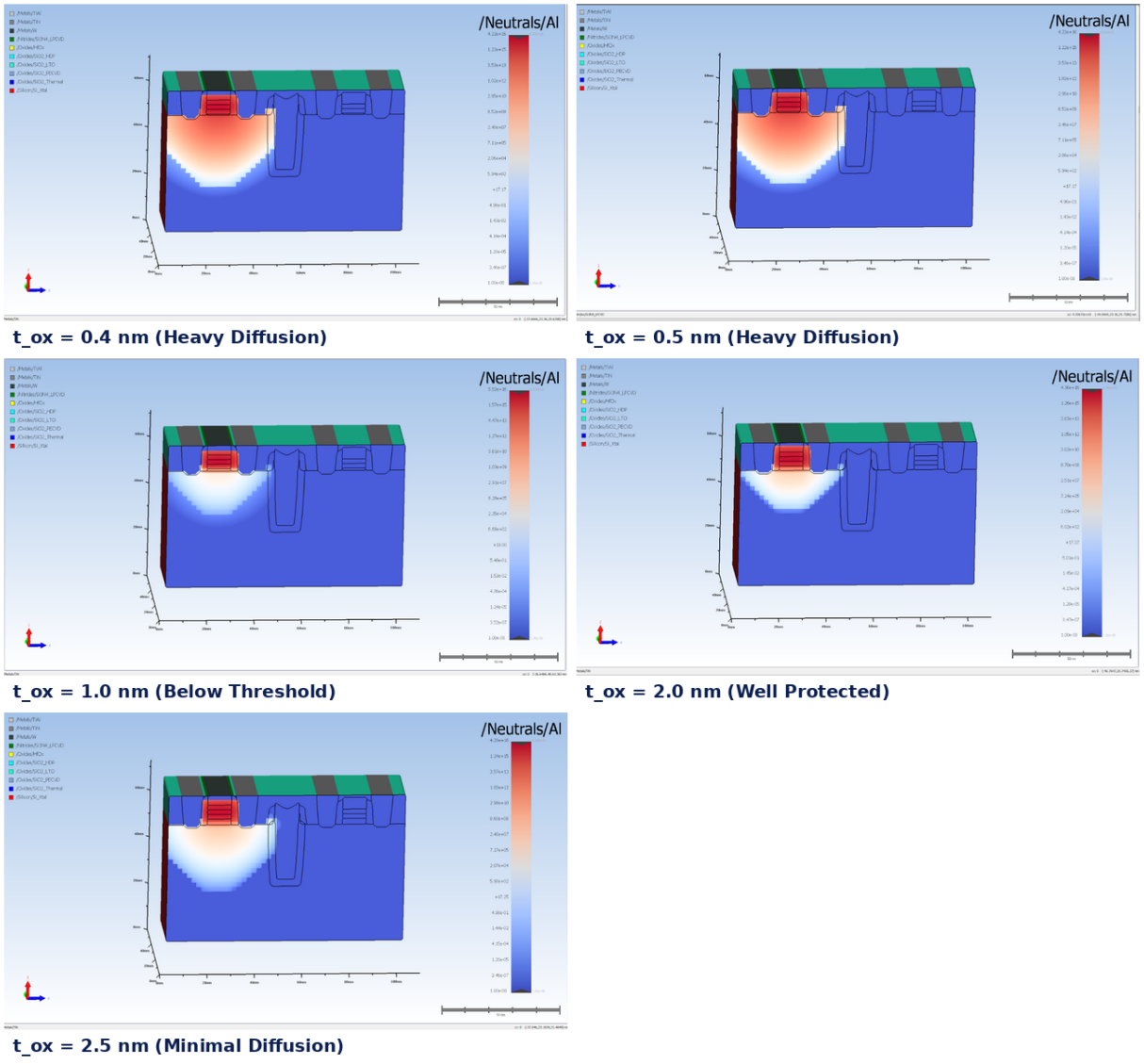


Figure 6 : Cross-Sectional Al Diffusion Profiles at Varying Gate Oxide Thicknesses (0.4 nm to 2.5 nm)

4. QUESTION 2: GATE LAST PROCESS ANALYSIS (ASSIGNMENT4_GATELAST.ZAM)

The Assignment4_GateLast.zam project was built and analyzed in Semulator3D. In this process flow, the gate stack is deposited AFTER the high-temperature anneal steps for the source and drain.

(a) Al Metal Diffusion After Dummy Anneal:

The sacrificial polysilicon dummy gate is refractory and does not diffuse during source/drain activation. In this project, a dummy anneal was added at the end to represent the cumulative thermal exposure of subsequent process steps. Under the standard dummy anneal at 1000°C, the simulated Al concentration in the active channel is only 1.64 atoms/cm^3, which is virtually zero compared to the 10^{12} atoms/cm^3 threshold.

Table 4 summarizes the complete Process Editor sequence for the Gate-Last (RMG) integration scheme.

Table 4: Process Editor Recipe Steps for Gate-Last (RMG) Flow (Assignment4_GateLast.zam)

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
1	Wafer Setup	Wafer Setup	Crystalline Si	100	
2	Conformal Deposit	Pad Oxide Deposition	SiO_2 (Thermal)	5	
3	Conformal Deposit	SiN hardmask	Si_3N_4 (LPCVD)	20	
4	Sequence	STI_Trench_Lithography			
4.1	Planarizing Deposit	Photo Resist coating	Photoresist	40	
4.2	Expose Material	Expose STI_Trench	Photoresist	0	STI_Trench
5	Sequence	Etching trench			
5.1	Enhanced Basic Etch	SiN hardmask etch	Nitrides, Organics, Oxides, Silicon	20	
5.2	Enhanced Basic Etch	pad oxide Etch	Nitrides, Organics, Oxides, Silicon	5	
5.3	Enhanced Basic Etch	substrate silicon Etch	Nitrides, Organics, Oxides, Silicon	50	
6	Remove Materials	resist removal	Organics, Photoresist		
7	Interface Growth	Thermal Oxidation of opened trench	Air/O_2, SiO_2 (Thermal), Silicon		
8	Conformal Deposit	Filling trench with CVD oxide	SiO_2 (PECVD)	20	
9	Polish	STI oxide CMP			
10	Enhanced Basic Etch	Removal of SiN hard mask	Si_3N_4 (LPCVD)	20	
11	Multi Etch (Advanced Modeling)	Pad_oxide ramoval	SiO_2 (PECVD), SiO_2 (Thermal)	5	
12	Sequence	Deposition_and_Patterning_of_HighK_Dielectric_and_Polygate			
12.1	Conformal Deposit	Thermal Oxide	SiO_2 (Thermal)	0.5	
12.2	Conformal Deposit	High-K_dielectric_deposition	HfO_x	2	
12.3	Conformal Deposit	Gate_Poly_Silicon_Deposition	Amorphous Si	20	

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
12.4	Planarizing Deposit	Planar Deposit	Photoresist	20	
12.5	Expose Material	Expose Material	Photoresist	0	gate
12.6	Enhanced Basic Etch	Basic Etch	HfO_x, Amorphous Si, Crystalline Si	30	
12.7	Enhanced Basic Etch	Basic Etch	Oxides	1	
12.8	Remove Materials	resist removal	Organics, Photoresist		
12.9	Conformal Deposit	Conformal Deposit	Si_3N_4 (LPCVD)	5	
12.10	Enhanced Basic Etch	Basic Etch	Si_3N_4 (LPCVD)	6	
12.11	Remove Materials	resist removal	Organics, Photoresist		
13	Sequence	Source_and_Drain_Formation			
13.1	Planarizing Deposit	Planar Deposit	Photoresist	20	
13.2	Expose Material	Expose Material	Photoresist	0	NMOS
13.3	Implant (Advanced Modeling)	Implant (Requires Advanced Modeling)			
13.4	Remove Materials	resist removal	Organics, Photoresist		
13.5	Planarizing Deposit	Planar Deposit	Photoresist	20	
13.6	Expose Material	Expose Material	Photoresist	0	PMOS
13.7	Implant (Advanced Modeling)	Implant (Requires Advanced Modeling)			
13.8	Remove Materials	resist removal	Organics, Photoresist		
13.9	Diffusion (Advanced Modeling)	Diffusion (Requires Advanced Modeling)	Metals, Al, TaN, TiN, HfO_x, SiO_2 (TEOS), Crystalline Si		
14	Sequence	ILD_Deposition_and_Polish			
14.1	Conformal Deposit	Conformal Deposit	SiO_2 (TEOS)	40	
14.2	Polish	STI oxide CMP			
15	Sequence	RMG_Pattern_Metal_1			
15.1	Planarizing Deposit	Planar Deposit	Photoresist	20	
15.2	Expose Material	Expose Material	Photoresist	0	layer1
15.3	Enhanced Basic Etch	Poly Pull Etch	Silicon	20	
15.4	Remove Materials	resist removal	Organics, Photoresist		
15.5	Conformal Deposit	Conformal Deposit	TiAl	2	
15.6	Planarizing Deposit	Planar Deposit	Photoresist	20	
15.7	Expose Material	Expose Material	Photoresist	0	NMOS
15.8	Enhanced Basic Etch	Basic Etch	TiAl, Si_3N_4 (LPCVD)	5	

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
15.9	Remove Materials	resist removal	Organics, Photoresist		
16	Sequence	RMG_Pattern_Metal_2			
16.1	Planarizing Deposit	Planar Deposit	Photoresist	20	
16.2	Expose Material	Expose Material	Photoresist	0	PMOS
16.3	Enhanced Basic Etch	Profile Etch	Crystalline Si	20	
16.4	Remove Materials	resist removal	Organics, Photoresist		
16.5	Conformal Deposit	Conformal Deposit	TiN	1	
16.6	Planarizing Deposit	Planar Deposit	W	20	
16.7	Polish	Stopping CMP			
17	Diffusion (Advanced Modeling)	Diffusion (Requires Advanced Modeling)	Nitrides, HfO_x, SiO_2 (TEOS), SiO_2 (Thermal), Crystalline Si		

4. GATE-LAST STRUCTURE AND PROCESS EDITOR IMAGES

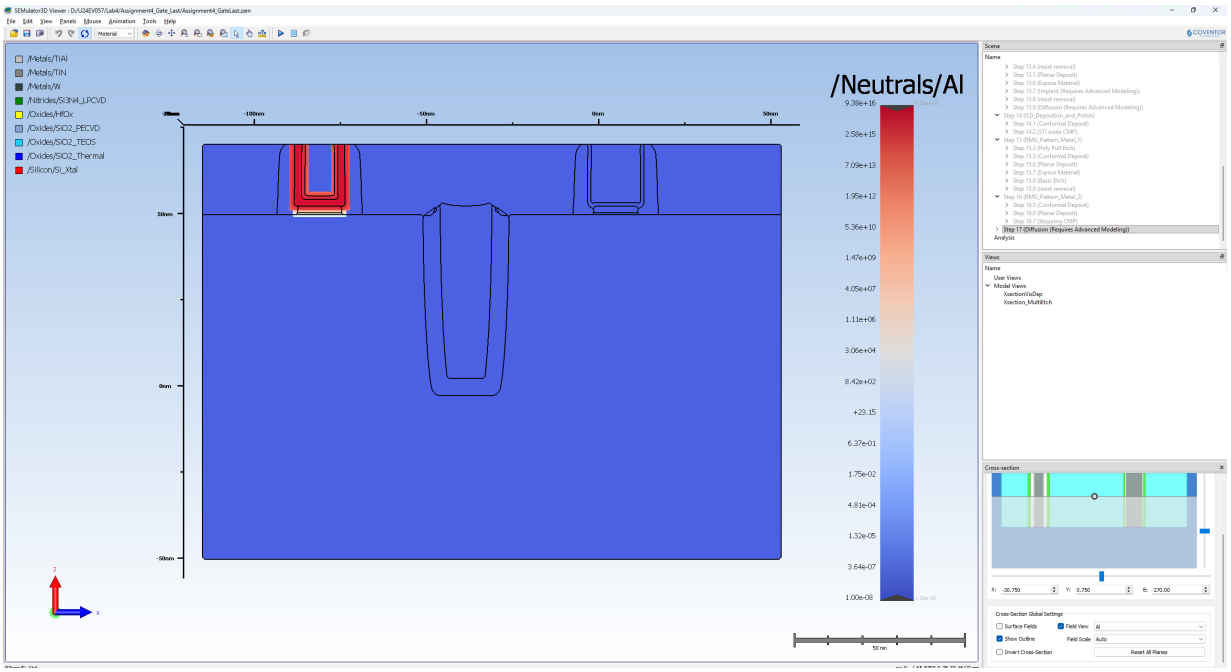


Figure 7 : Al Metal Diffusion in Gate-Last Structure

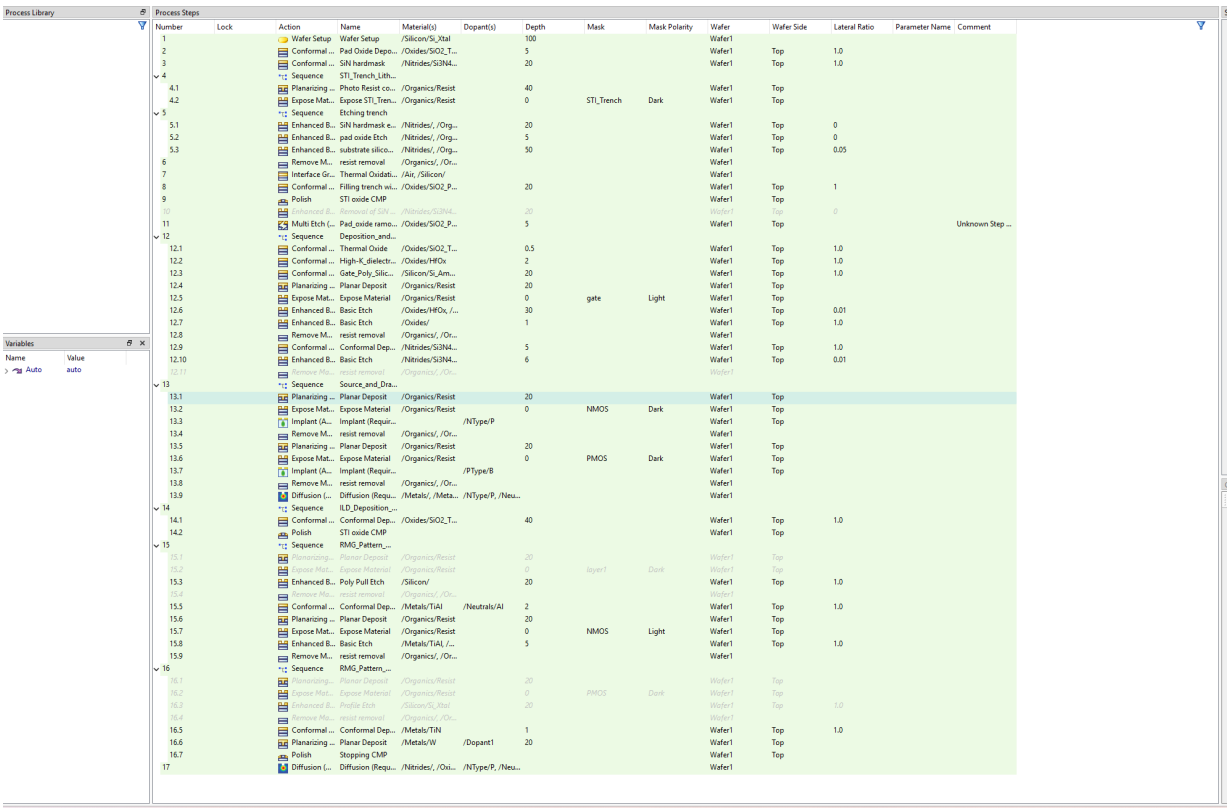


Figure 8 : Process Editor Steps for Gate-Last Flow

4. QUESTION 2 (B), (C) & (D): THERMAL MARGIN, COMPLEXITY & HAR CHALLENGES

(b) Maximum Dummy Anneal Temperature:

To evaluate the thermal margin of the replacement gate, the dummy anneal temperature was increased systematically from 900°C up to 1300°C. Contamination levels remained strictly below the critical failure limit of 10^{12} atoms/cm^3 across all simulated temperatures, reaching only 6.20×10^{10} atoms/cm^3 at 1300°C. Table 5 details the simulated contamination vs. dummy anneal temperature.

• Maximum temperature before contamination exceeds 10^{12} cm^{-3}: Safe up to 1300°C (Process remains compliant).

(c) Added Steps and Materials in Gate-Last (RMG) Flow:

The Gate-Last process solves the thermal budget and diffusion problems, but at the expense of added process complexity compared to Gate-First. Extra steps include sacrificial dummy gate deposition, ILD CMP, selective dummy poly pull etch, conformal ALD work-function metal deposition, bulk metal fill, and metal CMP.

Extra materials include sacrificial amorphous silicon (Si_Amorph), TEOS-based SiO_2 ILD, CMP stop liners, and dual work-function metal stacks (TiN, TaN, TiAl, TiAlC).

(d) High-Aspect-Ratio Metal Deposition and Scaling Limit:

A critical requirement of the metal-last process is high-aspect-ratio (HAR) metal deposition. As channel length decreases while gate height is held constant, the aspect ratio (Height / Length) increases significantly. At ultra-narrow gate lengths, precursor pinch-off at the trench opening leads to void formation and incomplete fill.

Table 5: Simulated Al Contamination Concentration vs. Dummy Anneal Temperature (Gate-Last Flow)

Anneal Temperature (°C)	Contamination Concentration (cm^{-3})	Status
900	4.00×10^2 (400)	Below Limit (Passes)
1000	5.97×10^5	Below Limit (Passes)
1200	2.50×10^9	Below Limit (Passes)
1250	1.32×10^{10}	Below Limit (Passes)
1275	2.90×10^{10}	Below Limit (Passes)
1300	6.20×10^{10}	Below Limit (Passes - Max Safe Temp Tested)

Table 6: Comparison between Gate-First and Gate-Last (Replacement Metal Gate) Flows

Category	Gate-First Flow	Gate-Last (Replacement Gate) Flow	Remarks / Impact
Gate Stack Deposition	Deposited before S/D high-temp anneal	Sacrificial poly gate first; metal gate deposited after S/D anneal	Eliminates metal thermal degradation
Thermal Budget Constraint	Severe (Metal must withstand > 1000°C)	High (Metal deposited after high-temp steps)	Allows lower-work-function & low-k materials
Extra Process Steps	None (Conventional self-aligned gate)	Sacrificial poly deposition, ILD CMP, Poly pull etch, Metal fill, Metal CMP	Higher lithography & CMP complexity
Additional Materials	Standard gate metals (TiN, TiAl, W)	Sacrificial Polysilicon, Sacrificial Oxide, ILD Oxide, Polish stop layers	Requires highly selective etch chemistries
High-Aspect-Ratio Challenge	None (Gate patterned subtractively)	Critical (Must void-free fill narrow trenches)	Requires atomic layer deposition (ALD)
Threshold Voltage Stability	V_{th} shift due to metal diffusion & reaction	High V_{th} stability and precise work function tuning	Standard choice for ≤ 14 nm logic nodes

5. QUESTION 3: INBUILT 14 NM FINFET PROCESS ANALYSIS (14NMFINFET.ZAM)

The industrial 14 nm FinFET process project (14nmFinFET.zam) was examined in Semulator3D to analyze gate stack integration in modern 3D multi-gate transistors.

(a) Identification of Gate Deposition Steps:

In the 14nmFinFET.zam process flow, the gate module is implemented as follows:

• Step 2.5.1 (DEP SacOx): Conformal deposition of sacrificial gate oxide over the patterned 3D silicon fins.

• Step 2.5.3 (DEP SacGate): Deposition of sacrificial polysilicon dummy gate filling between the fins (Figure 9).

• Step 2.5.4 (CMP SacGate): Planarization of the sacrificial dummy polysilicon gate.

• Step 2.5.10–2.5.19: Lithographic patterning and anisotropic etching of the dummy gate lines.

• Step 2.6.3 (RTA): Source/drain dopant activation anneal conducted with the dummy gate in place (Figure 10).

• Subsequent Steps: Dummy poly pull etch followed by dual work-function replacement metal gate ALD deposition.

(b) Process Flow Classification:

The 14 nm FinFET process is a Gate-Last (Replacement Metal Gate / RMG) architecture. The dummy poly gate acts as a place-holder during high-temperature source/drain epitaxy (SiGe for PMOS, SiC for NMOS) and RTA, protecting channel integrity.

(c) Gate Metal Materials for PMOS and NMOS:

Advanced FinFET CMOS requires dual work-function metal gate stacks to set low and symmetric threshold voltages (V_{th}) without excessive channel doping:

• NMOS Gate Metal: TiAl (Titanium Aluminum) or TiAlC — A low work-function metal (~4.1 to 4.3 eV) energetically aligned near the silicon conduction band (E_c).

• PMOS Gate Metal: TiN (Titanium Nitride) or TaN — A high work-function metal (~4.9 to 5.1 eV) energetically aligned near the silicon valence band (E_v).

5. 14 NM FINFET SACRIFICIAL GATE & POST-RTA STRUCTURE IMAGES

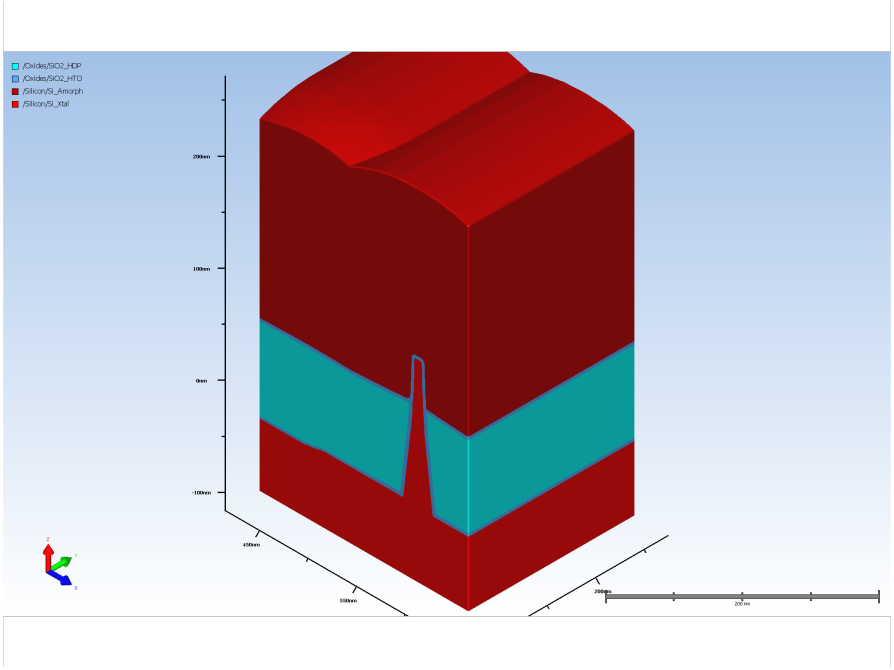


Figure 9 : Sacrificial Gate Deposition in 14 nm FinFET (Step 2.5.3)

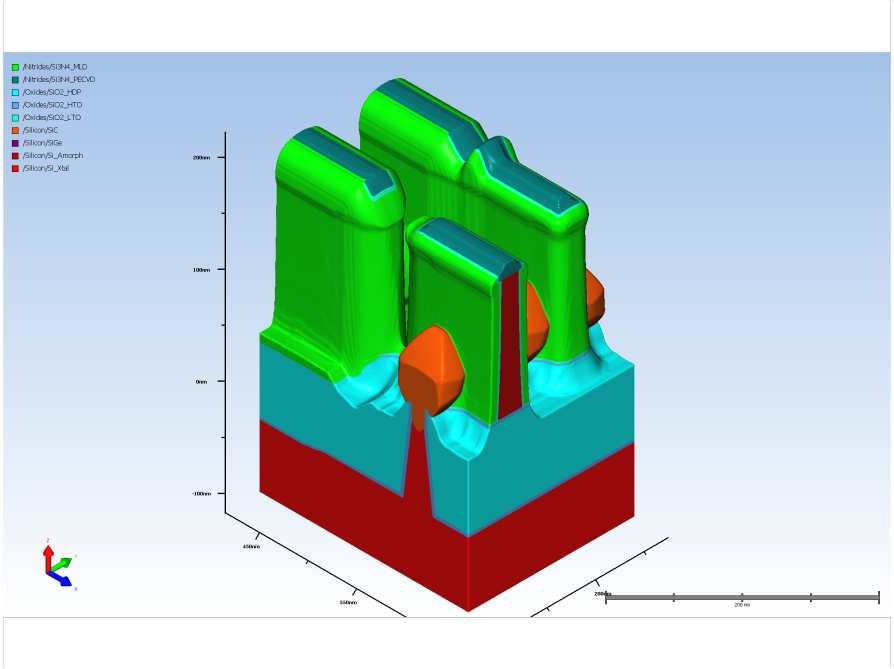


Figure 10 : 14 nm FinFET Structure Post-RTA Prior to RMG (Step 2.6.3)

6. OBSERVATIONS & CONCLUSION

Observations:

1. In Gate-First CMOS, the default 12 s activation anneal caused severe aluminum diffusion into the active silicon channel (1.24×10^{15} atoms/cm^3), exceeding the 10^{12} atoms/cm^3 operational limit.

2. Reducing the anneal duration suppressed metal contamination, establishing a maximum safe anneal time of approximately 6.14 seconds.

3. Increasing gate oxide thickness demonstrated that SiO_2 acts as an effective diffusion barrier; thicknesses ≥ 0.818 nm successfully suppressed contamination below 3.89×10^3 atoms/cm^3.

4. In Gate-Last CMOS, the refractory polysilicon dummy gate completely shields the active region during thermal anneals. Simulated contamination remained well below the 10^{12} atoms/cm^3 limit across all temperatures up to 1300°C (reaching only 6.20×10^{10} atoms/cm^3), verifying exceptional thermal margin.

5. Gate-Last flows achieve high thermal resilience at the expense of higher complexity: requiring sacrificial gate patterning, ILD CMP, selective poly pull, and high-aspect-ratio metal deposition.

6. Analysis of the 14 nm FinFET process confirmed the universal industry adoption of Gate-Last (RMG) architectures, utilizing TiAl for NMOS and TiN for PMOS to achieve symmetric threshold voltages.

Conclusion:

The trade-offs between Gate-First and Gate-Last integration schemes were evaluated using Semulator3D. While Gate-First suffers from severe thermal budget limitations due to metal cross-diffusion during source/drain activation anneals, the Gate-Last (RMG) flow decouples high-temperature anneals from gate stack deposition, enabling low-leakage, high-performance nano-scale CMOS devices.